CM3121 – Advanced Organic Chemistry

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Sigmatropic rearrangements

Cycloaddition reactions

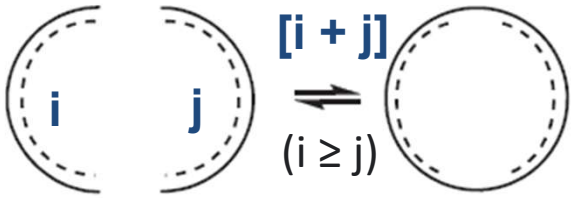
- A diverse range of ring forming reactions
- Pericyclic mechanism
- Frontier molecular orbitals and Woodward-Hoffmann rules
  - Thermal: total number of π electrons (4n+2) allowed, e,g. [4+2]
    Special case: thermal [2+2] (suprafacial-antarafacial)
  - Photochemical: total number of π electrons (4n) allowed, e.g. [2+2]
- Stereospecific
- Stereoselective
- Regioselective
- Catalysis (Lewis acid, hydrogen bonding interactions)
- Applications
  - Synthesis of functionalized, cyclic structures of various sizes
  - Bio-conjugation

n = a whole number (0, 1, 2, 3…)

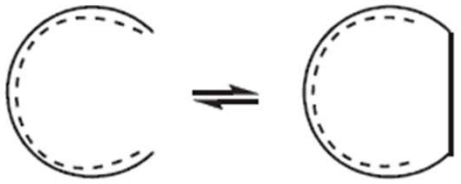
Pericyclic reactions (classifications)

- Cycloadditions
- Electrocyclic reactions
- Sigmatropic rearrangements
  - [3,3]-Sigmatropic rearrangements
  - [2,3]-Sigmatropic rearrangements
  - [1,n]-Sigmatropic hydrogen shifts

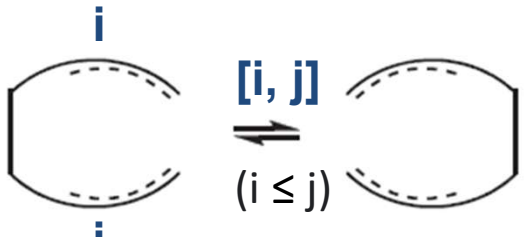
[i, j]: number of atoms the bond moved across (including both ends) the π systems



(a) Cycloadditions

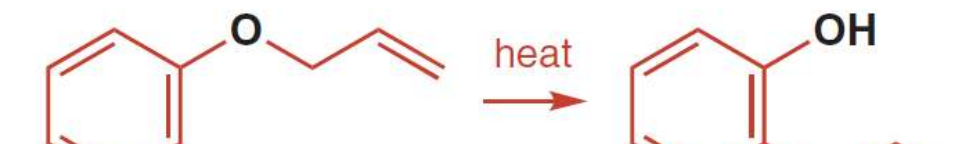


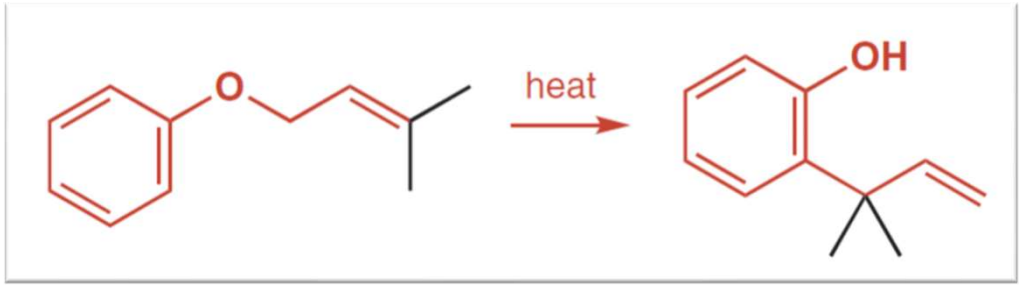
(b) Electrocyclic reactions



(c) Sigmatropic rearrangements

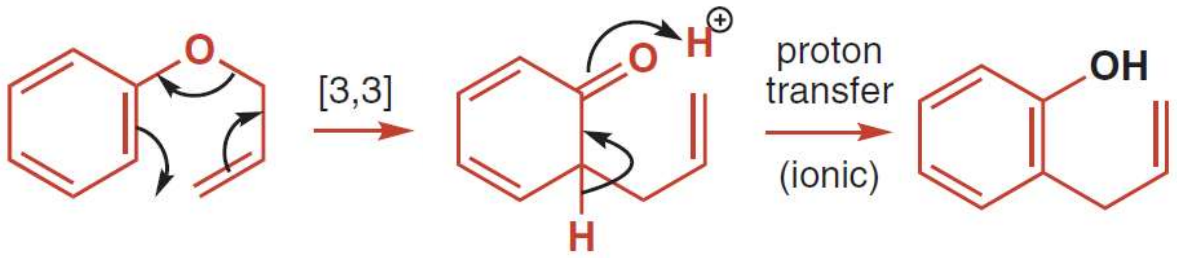
1. The Claisen rearrangement





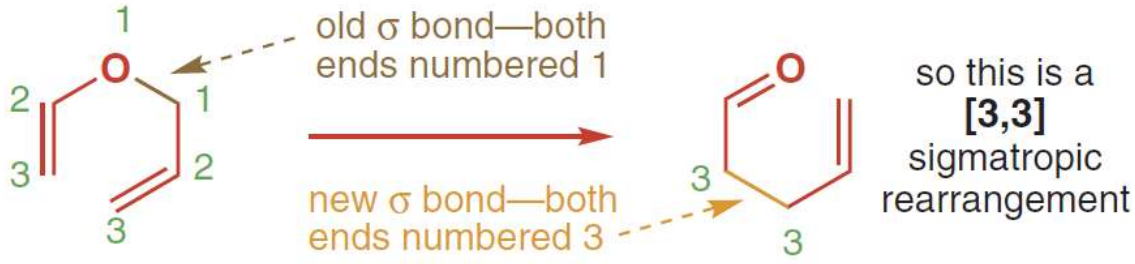
How to explain the regioselectivity of the product?

Proposed mechanism: [3,3]-sigmatropic rearrangements



The Claisen rearrangement

A type of [3,3]-sigmatropic rearrangements



Single Bond Energies (kJ/mol of bonds)

	H	C	N	O	S	F	Cl	Br	I
H	436								
C	413	346							
N	391	305	163						
O	463	358	201	146					
S	347	272	—	—	226				
F	565	485	283	190	284	155			
Cl	432	339	192	218	255	253	242		
Br	366	285	—	201	217	249	216	193	
I	299	213	—	201	—	278	208	175	151

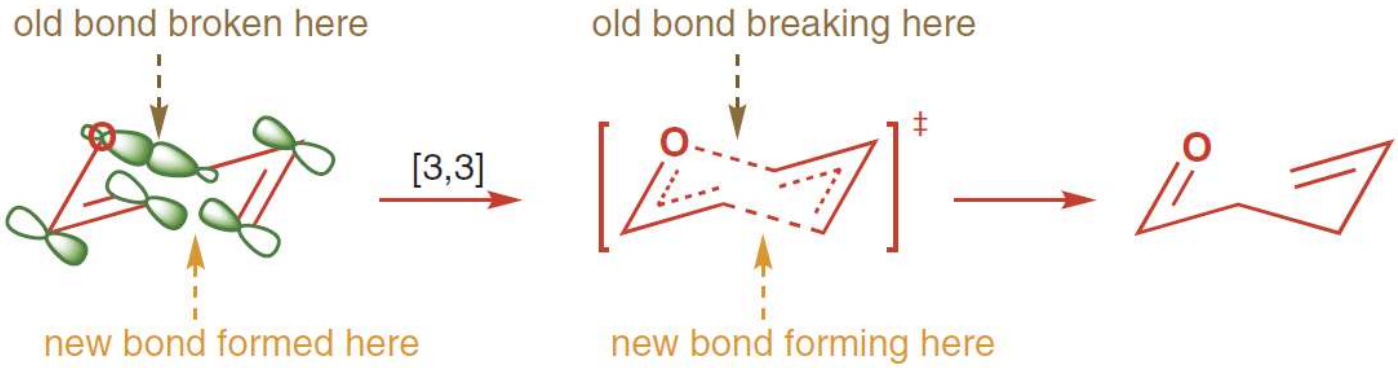
Thermodynamics: the product is favoured because of the high bond energy of C=O.

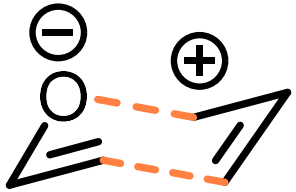
Multiple Bond Energies (kJ/mol of bonds)

C=C 602	C=N 615	C=O 799
C≡C 835	C≡N 887	C≡O 1072
N=N 418	N=O 607	
N≡N 945	O=O 498	

Mechanism: a chair-like transition state

The diagram only shows the orbitals involved in the reaction, not the frontier orbitals (HOMO LUMO). It does not tell us if the reaction is allowed.

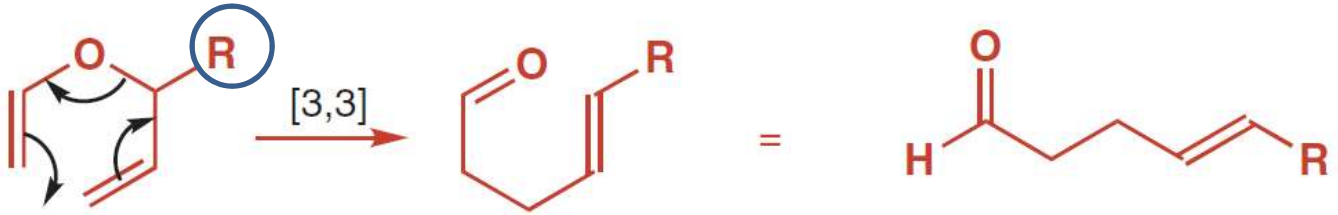


Consider the HOMO-LUMO interactions between  (suprafacial-suprafacial)

Why the **chair-like transition state** is allowed according to the Woodward–Hoffman rules?

In a thermal pericyclic reaction the total number of (4q + 2)s and (4r)a components must be odd.

Stereoselectivity: Chair-like transition state



γ,δ-unsaturated carbonyl compounds

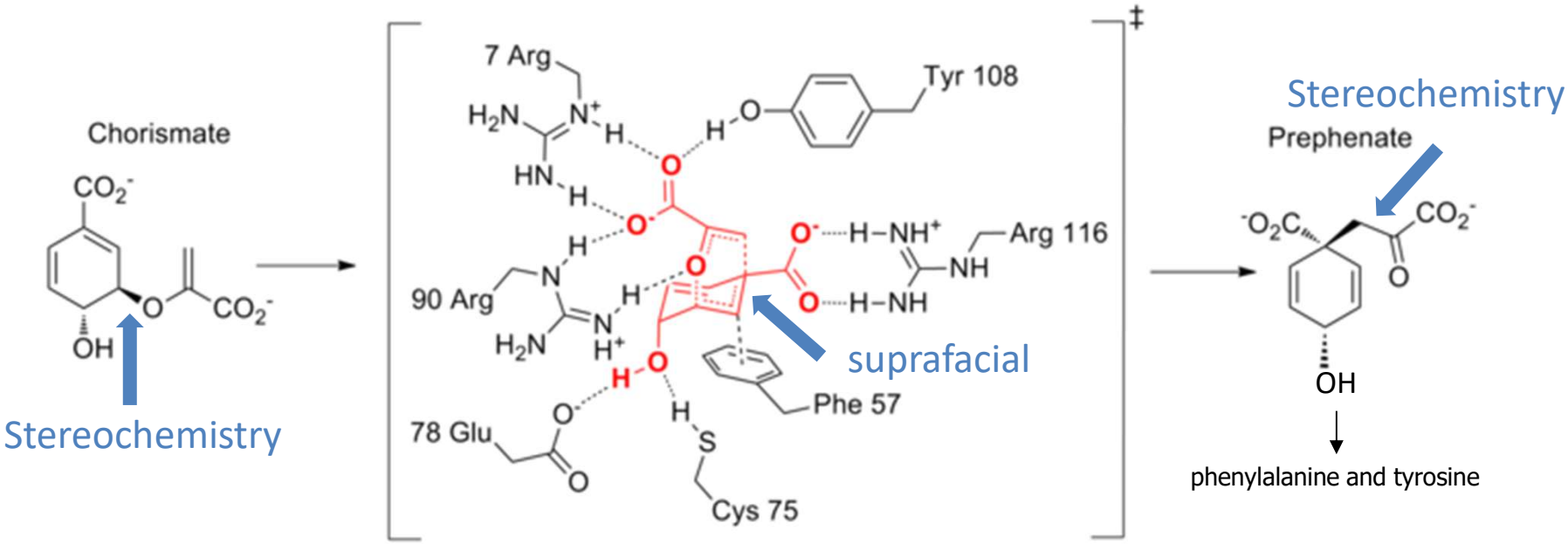


chair-like transition state

trans (*E*) geometry

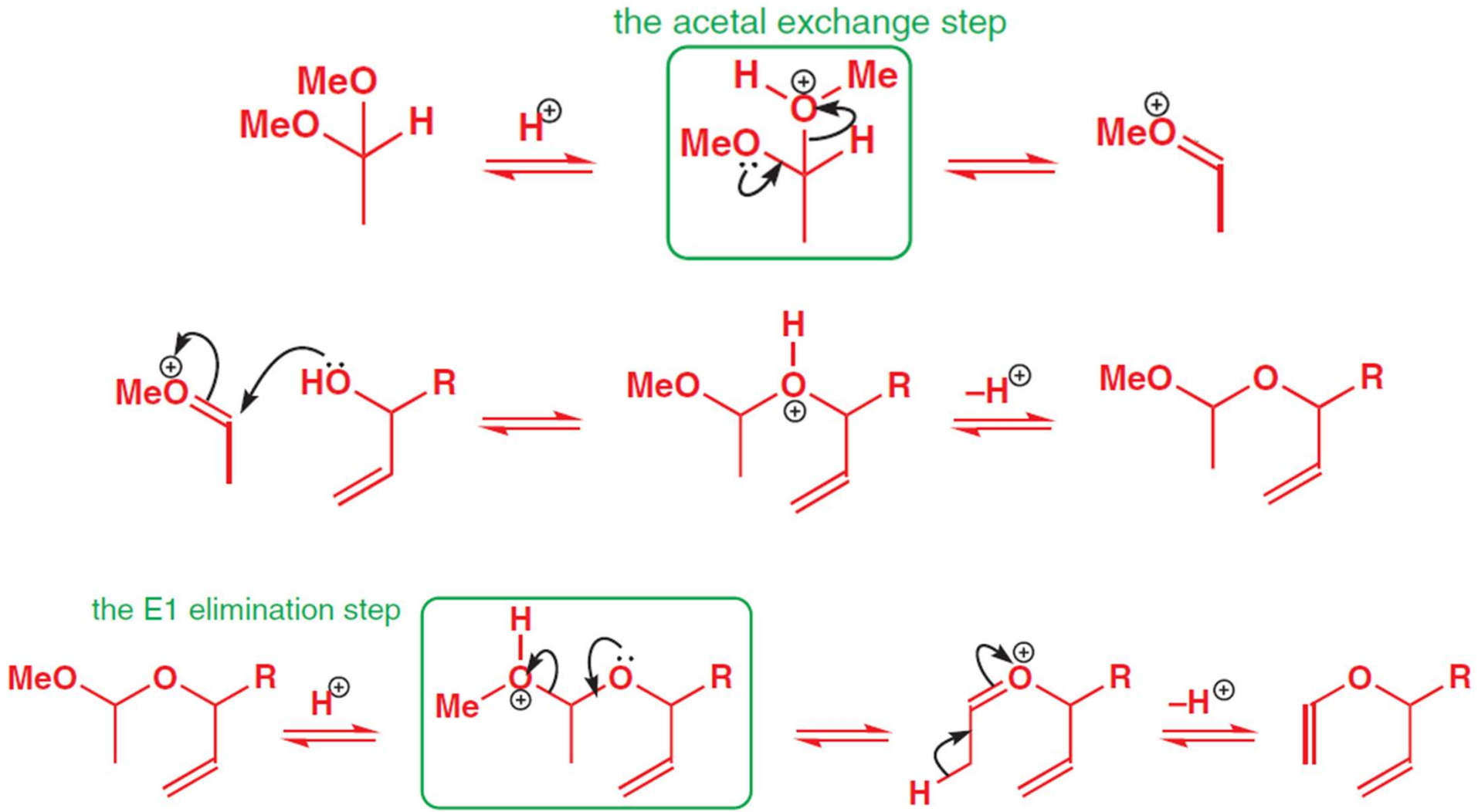
Claisen rearrangement in Nature
Pericyclic reactions (features)

- Transition state possesses a cyclic geometry
- No new radical or ionic intermediate within the cyclic transition state
- Bond-making and bond-breaking progress in a concerted fashion
- Stereospecific and/or stereoselective

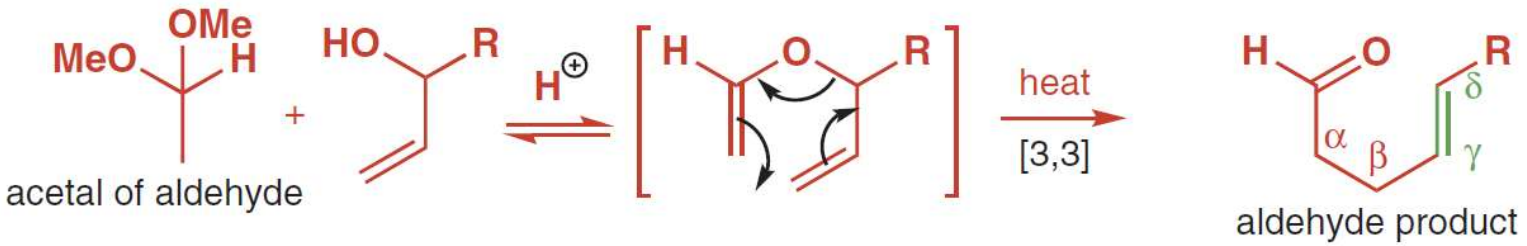


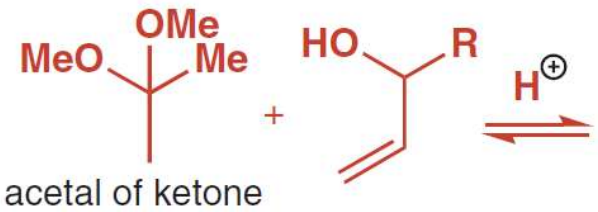
Chorismate mutase catalyzes the Claisen rearrangement of chorismate

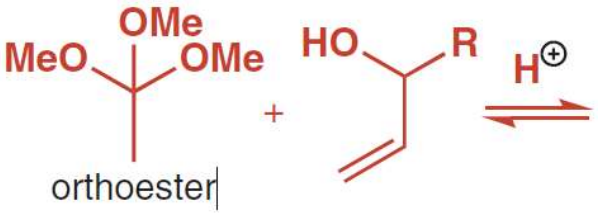
Formation of vinyl ether from acetal and alcohol

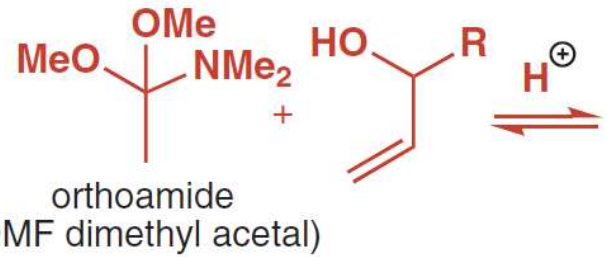


Synthesis of γ,δ-unsaturated carbonyl compounds



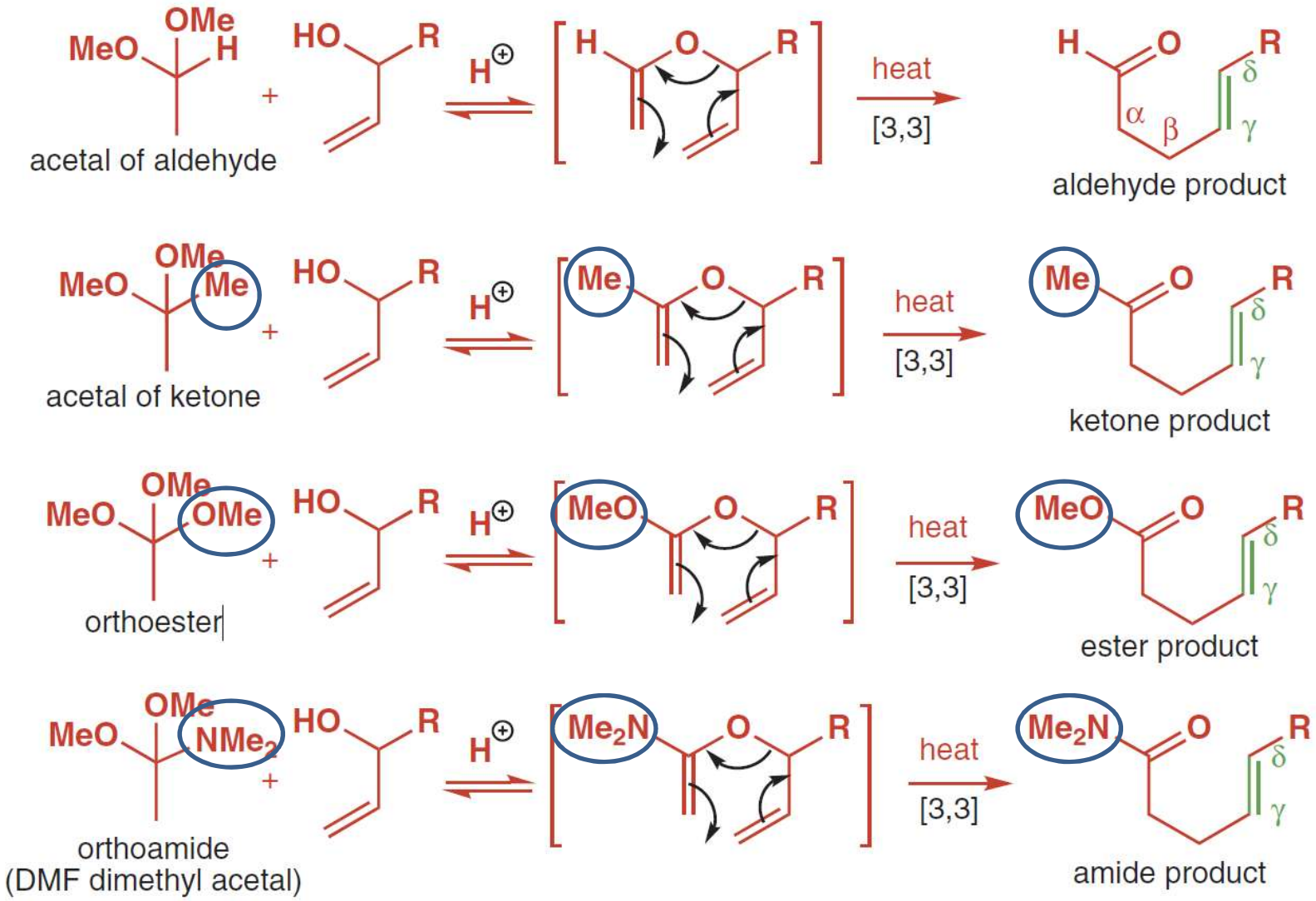






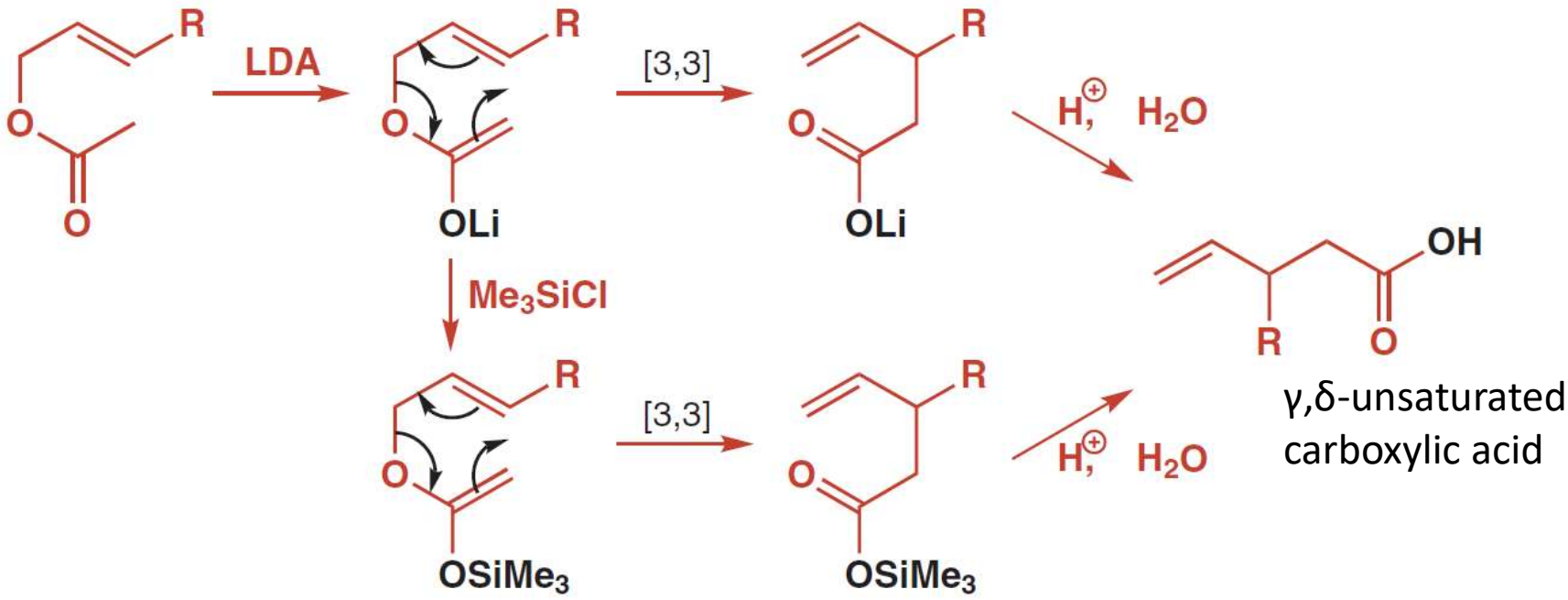
Proposed the structures of the [3,3] rearrangement products

Synthesis of γ,δ-unsaturated carbonyl compounds



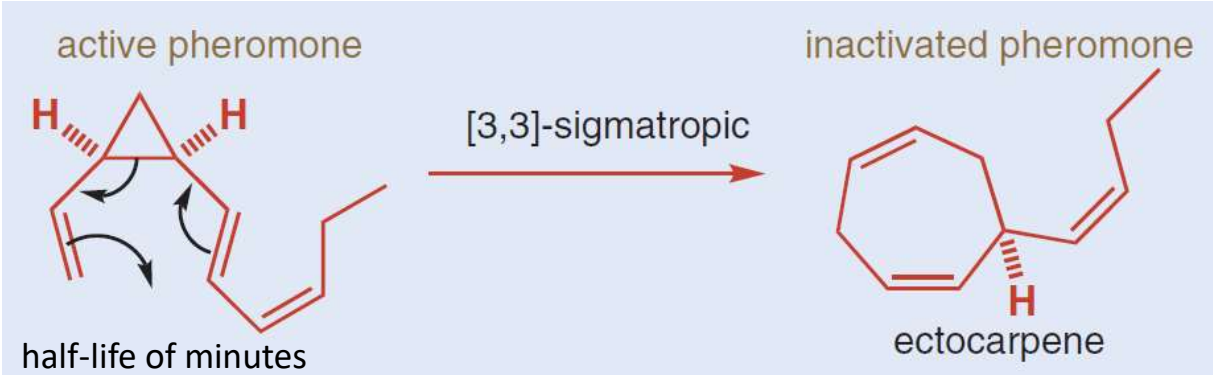
2. Ireland–Claisen rearrangement

- Combination of an oxygen atom in the ring and another one outside the ring



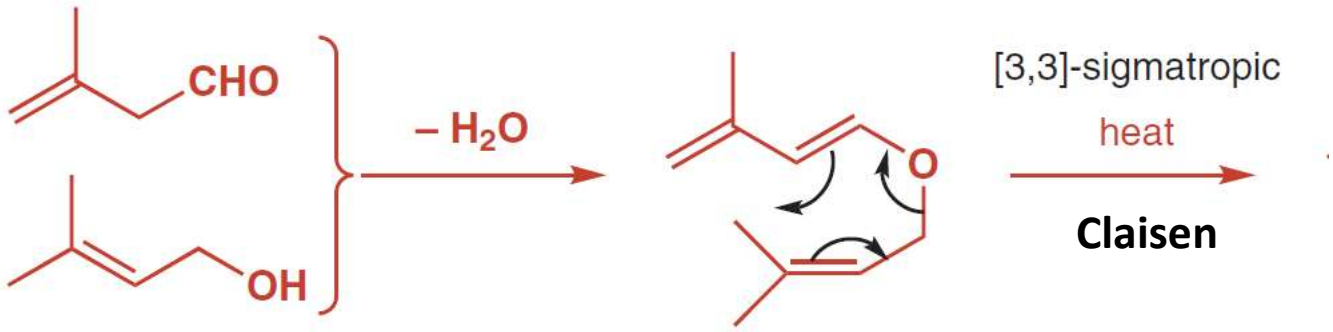
3. Cope rearrangement

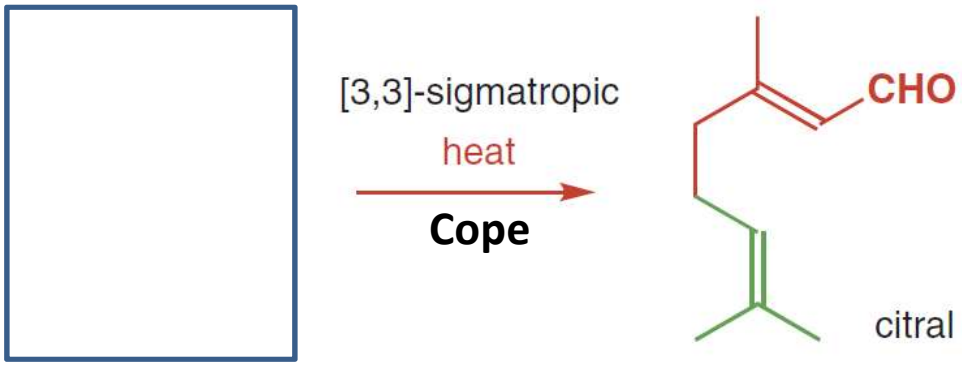




Driving force: release of strain from the three-membered ring

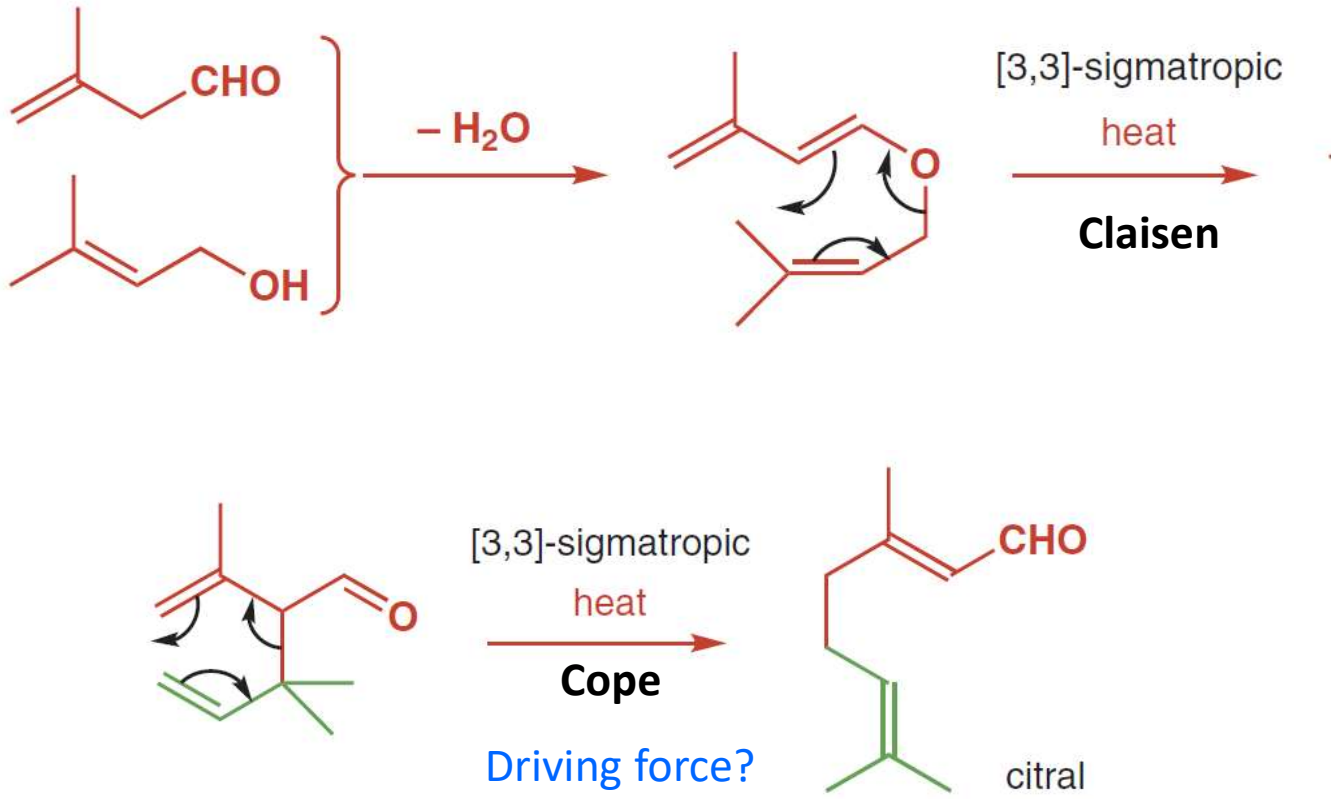
Industrial synthesis of citral





strong lemon odor

Industrial synthesis of citral

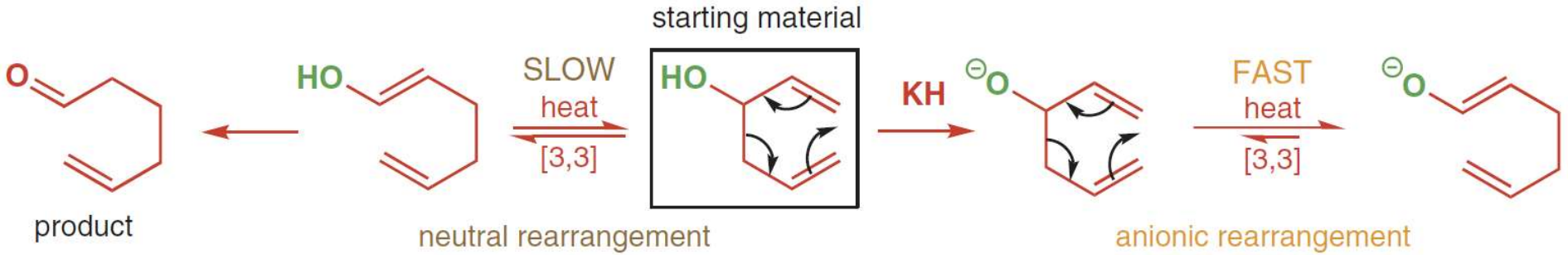


Driving force?

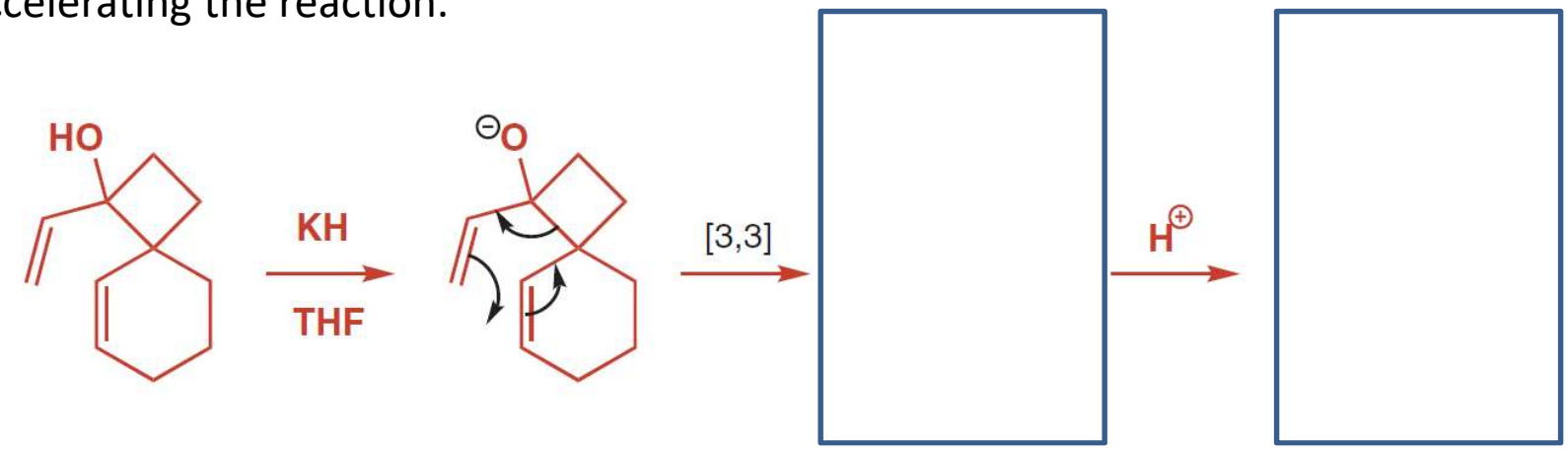
strong lemon odor

oxy-Cope rearrangement

Introducing 3-hydroxyl group

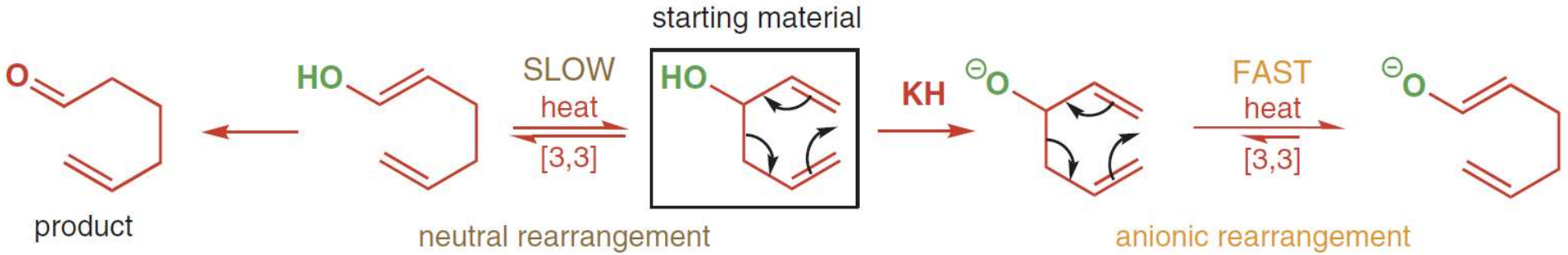


Conjugation between O and the new π bond stabilises the transition state, accelerating the reaction.

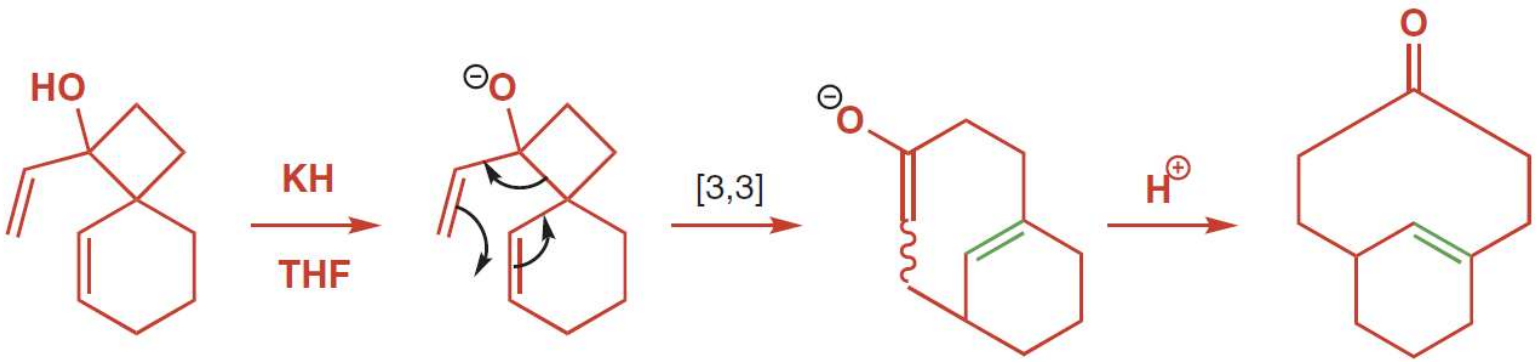


oxy-Cope rearrangement

Introducing 3-hydroxyl group

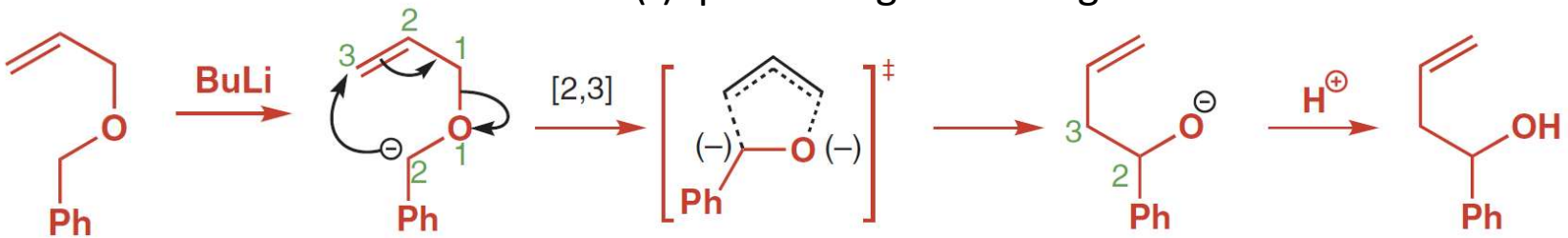


Conjugation between O and the new π bond stabilises the transition state, accelerating the reaction.



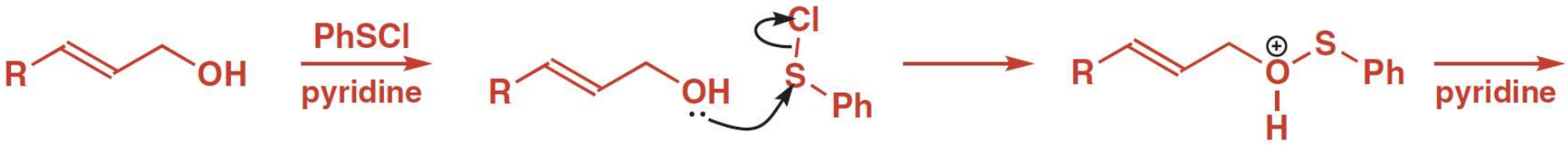
4. [2,3]-sigmatropic rearrangements with C(-)

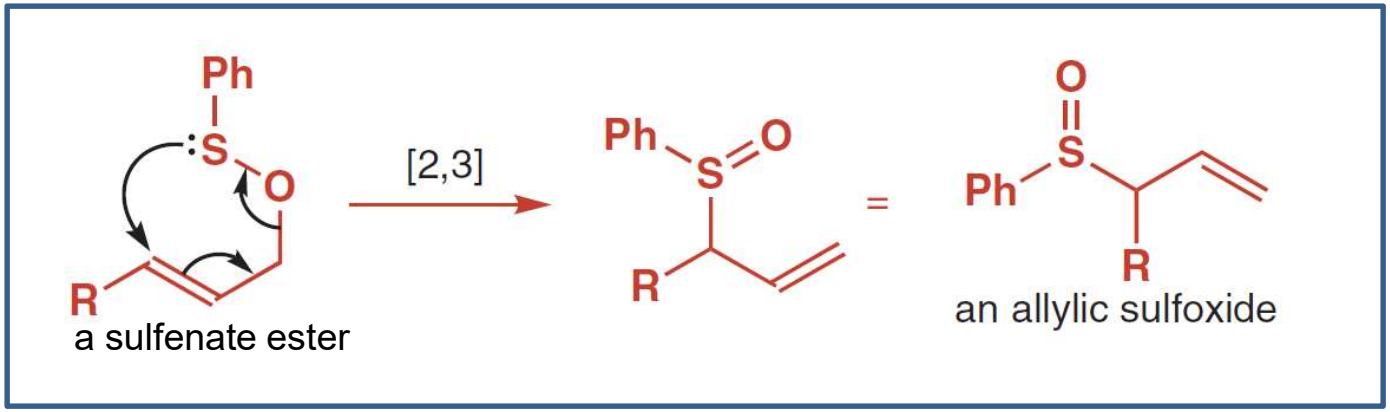
(-): partial negative charge



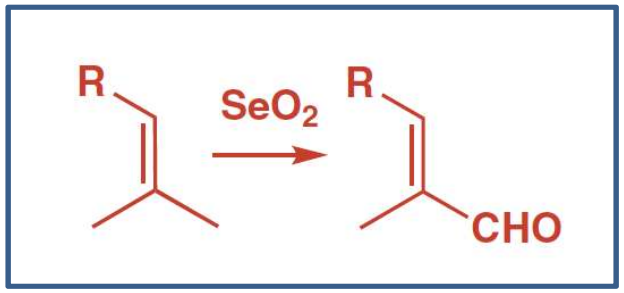
(suprafacial-suprafacial)

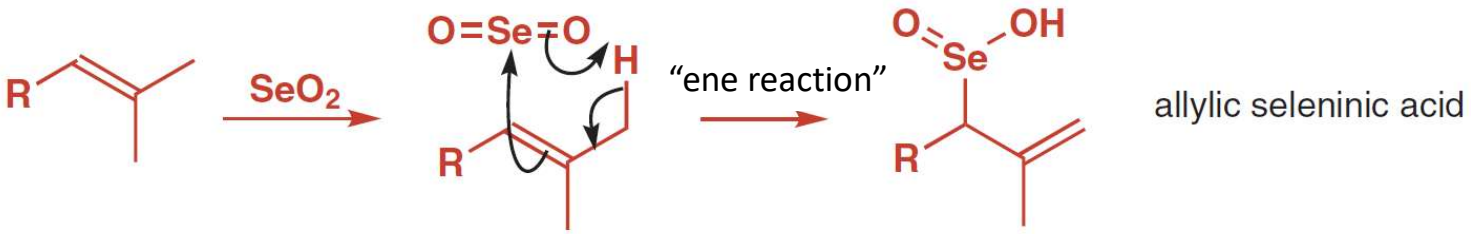
5. [2,3]-Sigmatropic rearrangements with S

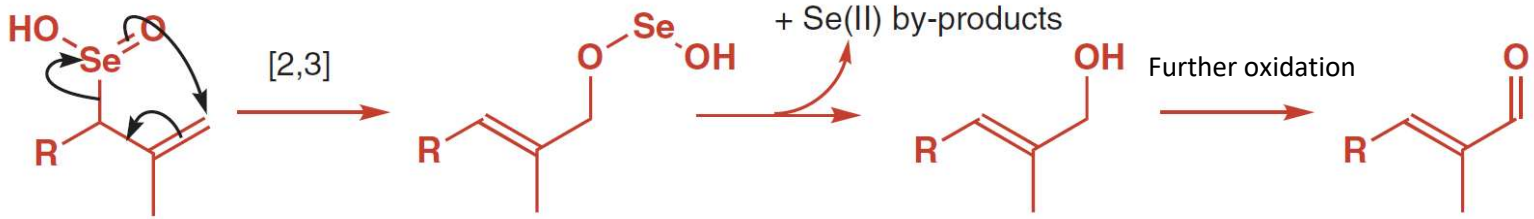




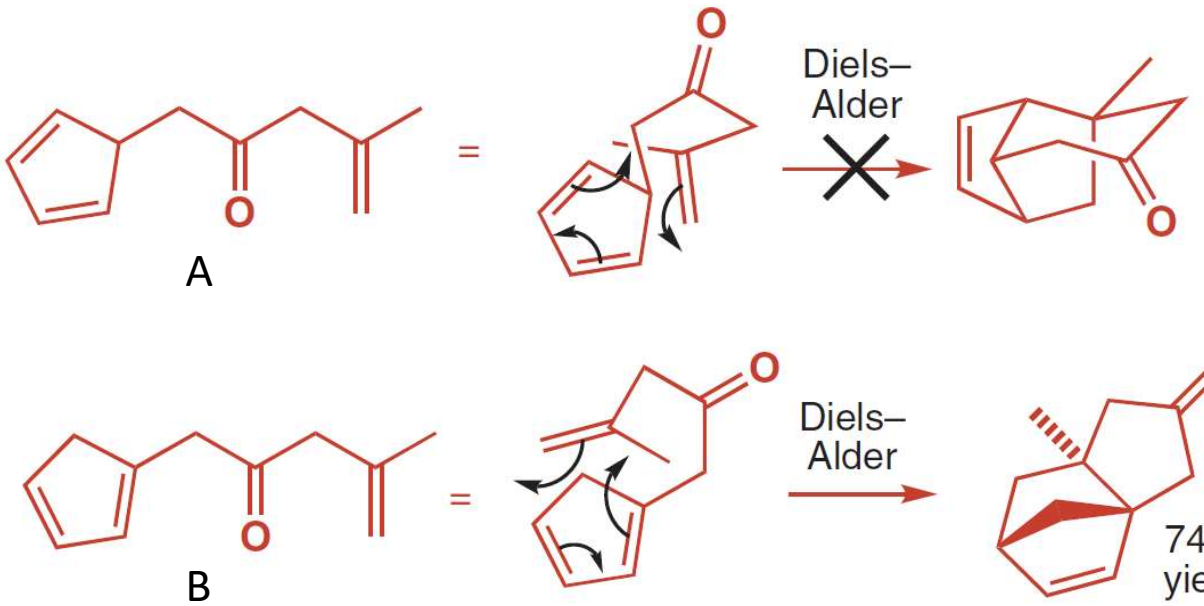
[2,3]-Sigmatropic rearrangements with Se



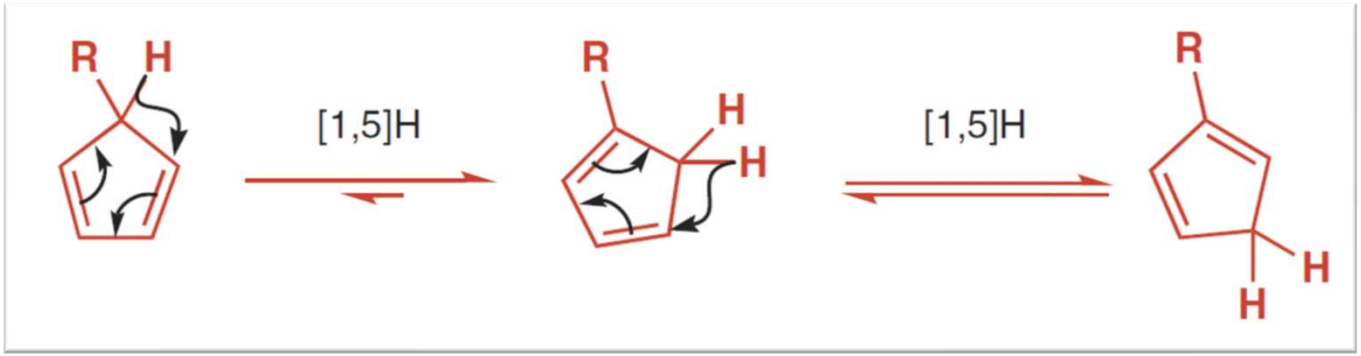




6. [1,5]-Sigmatropic hydrogen shifts

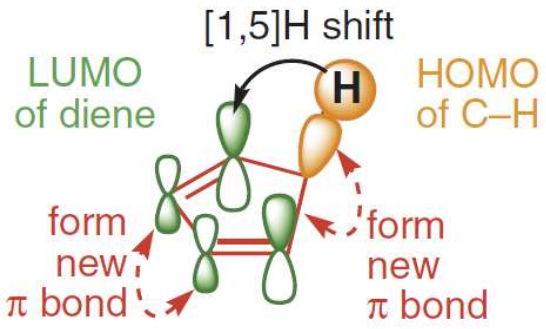


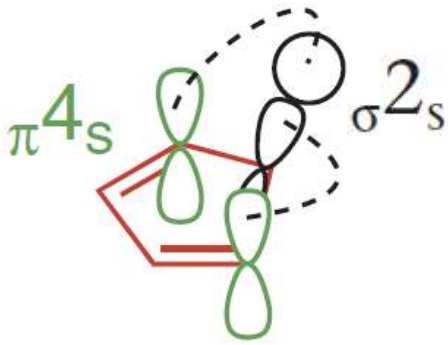
Same product regardless of using A or B as the starting material



Frontier orbital description and Woodward–Hoffman rules

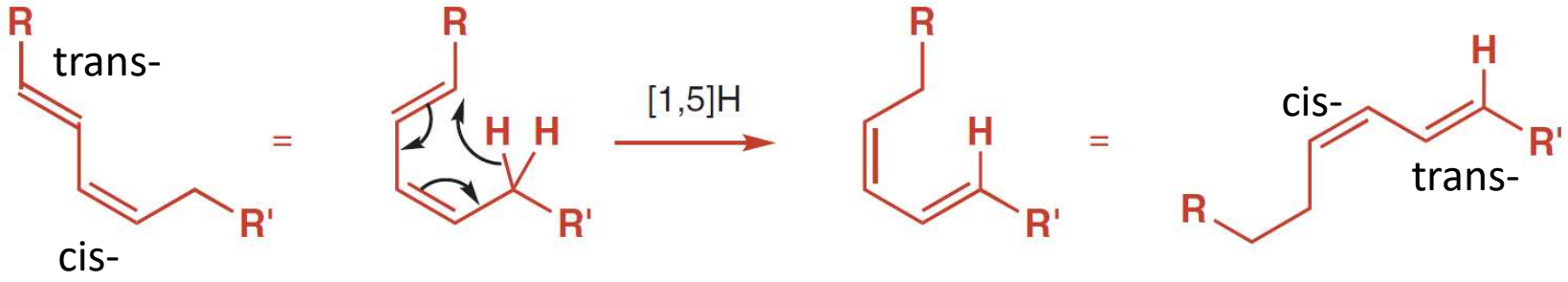
Suprafacial H migration





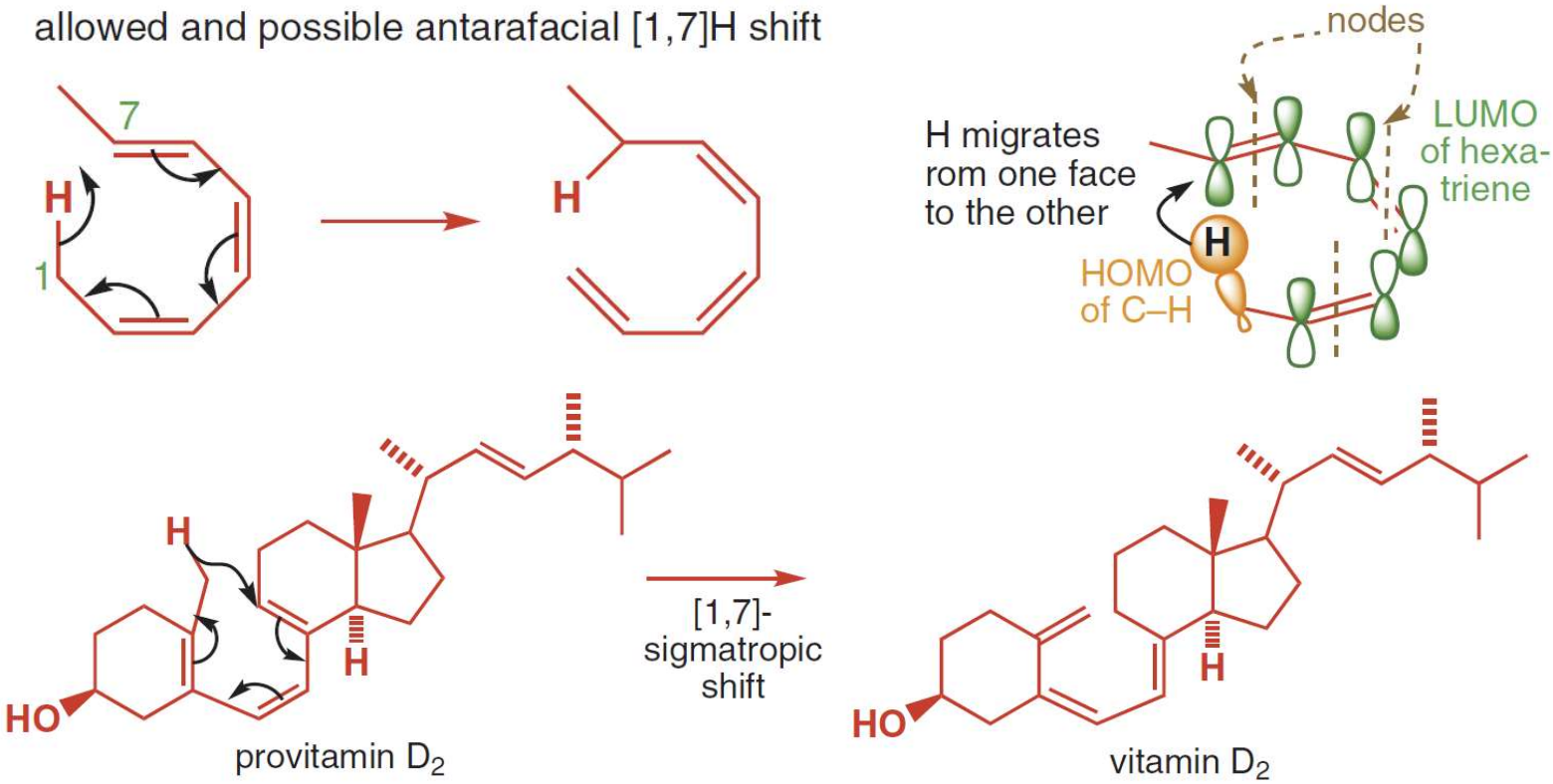
In a thermal pericyclic reaction the total number of (4q + 2)s and (4r)a components must be odd.

Make sure you join the specific orbital lobes that are going to form new bonds (dotted lines).

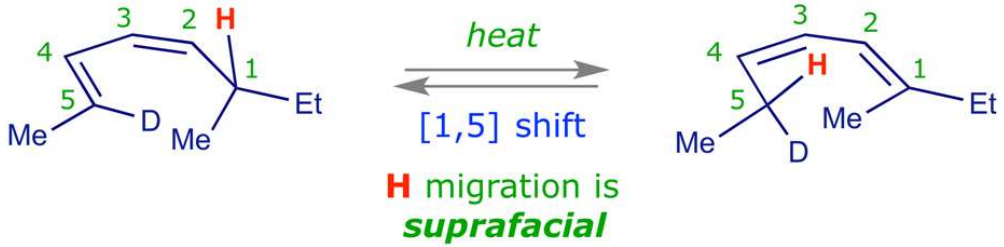


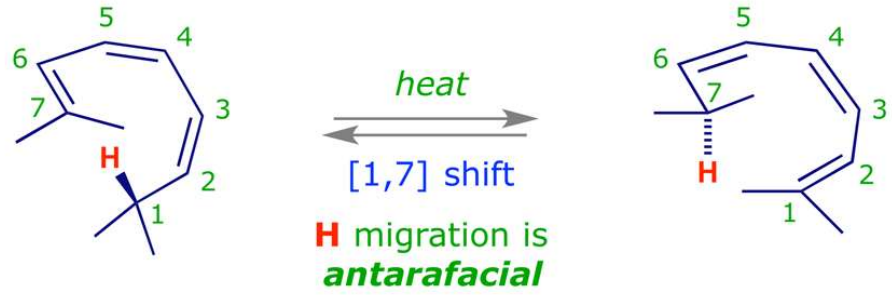
What about the E,E-diene (trans-trans)?

7. [1,7]-Sigmatropic hydrogen shifts (thermal: antarafacial)

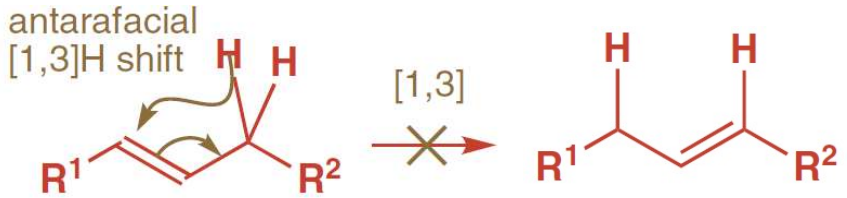


Thermal	[1,3]H shift	[1,5]H shift	[1,7]H shift
stereochemistry	antarafacial	suprafacial	antarafacial
feasibility	impossible	easy	possible





Why [1,3]H shifts do not take place?

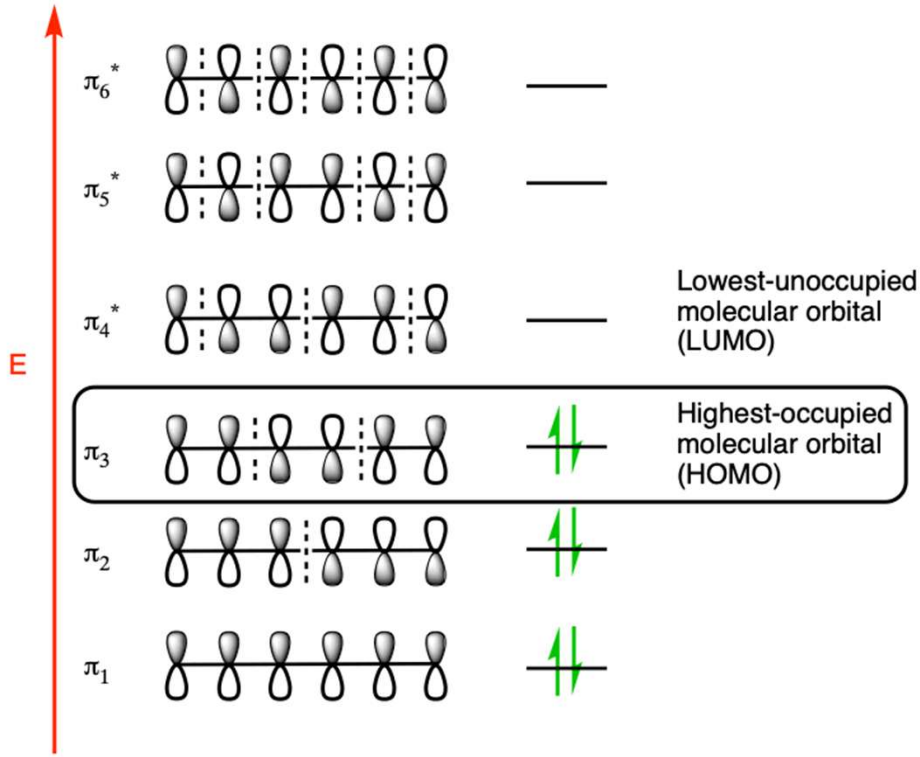


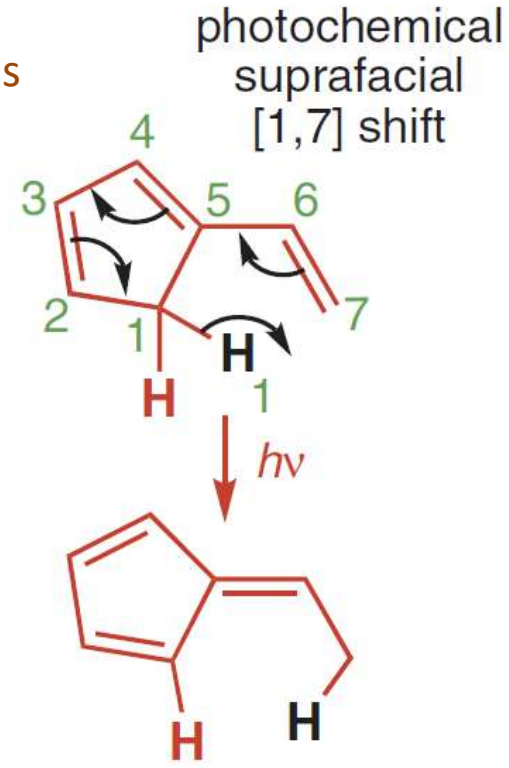
8. Photochemical [1,7] sigmatropic H shift (suprafacial)

- Photochemical and thermal reactions follow the opposite rules.

What are the molecular orbitals of excited state?

Photochemical conditions: HOMO(triene)-HOMO(C-H) interactions





(thermal [1,5] H shift is also possible)

Sigmatropic rearrangements

- [3,3]-Sigmatropic rearrangements
  - Claisen
  - Ireland–Claisen rearrangement
  - Cope
  - Oxy-Cope
- [2,3]-Sigmatropic rearrangements
  - C(-)
  - S
  - Se
- [1,n]-Sigmatropic hydrogen shifts
  - [1,5] H shift (thermal: suprafacial)
  - [1,7] H shift (thermal: antarafacial; photochemical: suprafacial)